Kyung (Chris) T. Han

truetheta@gmail.com

EDUCATION

University of Massachusetts Amherst Amherst, MA

Doctorate in Research and Evaluation Methods (Psychometrics), May 2008

Middle Tennessee State University Murfreesboro, TN

Graduate coursework in quantitative psychology (before transferring to UMass Amherst), 2003

Sung-Kyun-Kwan University Seoul, South Korea

B.A. in Business Administration, February 2000

Second Major: Science of Policy Administration

Minor: Education; Special Training: Military Science (ROTC)

HONORS, AWARDS, GRANTS, & PATENT

NCME Bradley Hanson Award for Contributions to Educational Measurement, 2024

US Patent (US 11,636,775 B2) on Honeycomb structure for automated pool assembly of test questions for test administration, 2023

Qiyas Award for Excellence in Research and Development, National Center for Assessment, Saudi Arabia, 2018

NCME Annual Award, 2017

AERA Div. D Significant Contribution to Educational Measurement Award, 2015 (on edited volume: Computerized Multistage Testing: Theory and Application; contributed two chapters)

NCME Jason Millman Promising Measurement Scholar Award, 2013

NCME Alicia Cascallar Award for Outstanding Paper by Early Career Scholar, 2012

Outstanding Graduate Student Grants, Applied Psychological Measurement, 2006

Graduate Council Travel Grant, University of Massachusetts Amherst, 2006

Outstanding Mission Performance Award, Republic of Korea Army, 2000

Best Literature Award for Mobile Internet Services ($5,000), KTF PCS Co., South Korea, 2000

Outstanding Business Administration Undergraduate Scholarship (full tuition), Sung-Kyun-Kwan University, 1996, 1997

EMPLOYMENT HISTORY

Graduate Management Admission Council (GMAC) June 2008 – Present

Senior Director, Head of Test Development and Psychometrics (TD&P) June 2017 – Present

Lead Department of Test Development and Psychometrics at GMAC (a) to design, develop, and launch assessment products to best serve the graduate management education market, (b) to design and implement advanced test drivers based on CAT, MST, LOFT, and (c) to conduct psychometric and operational research to continue to renovate and improve the measurement quality and test experience of existing test products including GMAT, NMAT, and EA.

Major achievements include: (1) designed and developed NextGen GMAT – based on 20+ industry-first inventions and CAT innovations rigorously reviewed by psychometric community, (2) developed GMAT Online, (3) developed Executive Assessment Online, (4) oversaw transition of LOFT-based NMAT exam to CAT, (5) designed and developed Executive Assessment based on new MST design with intersectional routing, (6) oversaw Template-based Automated Item Generation (AIG) platform development, (7) led LLM-based AIG platform development, (8) led LLM-based test development tools, and (9) designed and developed Honeycomb-based CAT item pool construction tools (US patented).

Research Director; Senior Psychometrician — March 2014 – May 2017

Led psychometric research initiatives, managed operational psychometric analyses, and oversaw item calibration and scoring processes for GMAT and Executive Assessment programs.

Psychometrician — March 2011 – February 2014

Associate Psychometrician — June 2008 – February 2011

Educational Testing Service — June 2007 – July 2007

Research Intern

Worked with Matthias von Davier (primary mentor), conducting studies on the general diagnostic model using TOEFL and GRE data. Also worked with Isaac Bejar (secondary mentor) on simulation tool developments.

Center for Educational Assessment, University of Massachusetts Amherst — September 2004 – May 2008

Senior Research Assistant

Supported the MST-based MAPT test development (adult education) with simulation and item scaling tool developments and the technical report.

Republic of Korea Army — March 2000 – June 2002

Personnel Officer (1st Lieutenant)

Handled all HR and budget-related responsibilities for a battalion of 400+ soldiers. Conducted a research project, 'Successful Leadership in a Battalion.'

ADVISORY & CONSULTANT EXPERIENCES

Law School Admission Council (LSAC) – Technical Advisory Committee member — 2025–present

World Bank – Measurement Expert Consultant — 2021–present

Korean Institution of Research for Behavior Science (KIRBS) – Technical Advisor — 2023–present

Educational Records Bureau (ERB) – Technical Consultant (via Sireci Psychometric Services) for operating CPT5 exam for K-12 educational outcomes — 2018–present

NextGen Adaptive CogAT development taskforce at Riverside Insights – Technical Advisory Committee Chair — 2024

Association for Measurement and Evaluation in Education and Psychology (AMEEP) – Scientific Committee member — 2024

Korea Health Personnel Licensing Examination Institute – Technical Advisor for test design — 2023

King Sejong Institute (under Ministry of Culture, Sports, and Tourism of Korea) – Co-investigator for development of Sejong Korean Language Assessment (SKA) — 2023–2024

Korea Institute for Curriculum and Evaluation (KICE) – Co-investigator for developing Multistage Test system for National Accountability Assessment — 2020–present

Universidad de Chile, Departamento de Evaluación, Medición y Registro Educacional – International Advisory Council member for SSAT — 2019–2020

Educational Records Bureau (ERB) – Senior test development expert (via Sireci Psychometric Services) for designing and developing MST-based CPT5 exam for K-12 educational outcomes 2016–2018

Educational Records Bureau (ERB) – Senior test development expert (via Sireci Psychometric Services) for developing CPT4R exam for K-12 educational outcomes 2015–2016

Korea Institute for Curriculum and Evaluation (KICE) – Advisor for National Accountability System development 2014–2019

Ministry of Education, Science and Technology, Korea – Technical Advisor for Enhanced Employability & Upgraded Proficiency (TEENUP) development 2012–2016

Jagiellonian University, Poland – Measurement Curriculum Specialist & Visiting Professor 2012–2014

Veebit – Research Committee Chair for development of Veebit Value Profiler Assessment kit 2013–2018

Korea Institute for Curriculum and Evaluation (KICE) – Principal Investigator for Assessment 2.0 2013–2014

Korea Institute for Curriculum and Evaluation (KICE) – Technical advisor for Automated Scoring System for Constructed Responses 2013–2016

Korea Institute for Curriculum and Evaluation (KICE) – Technical Advisor for NEAT score equating 2012–2013

COMMUNITY SERVICES

Editorial Service

Editorial board / manuscript reviewer (Journal of Computerized Adaptive Testing), 2012–2023

Editorial board (Journal of Educational Measurement), 2019–2025

Editorial board (Quantitative Psychology and Measurement), 2021

Software review editor (International Journal of Testing), 2009–2011

Manuscript & Report Review

Manuscript reviewer (Applied Psychological Measurement), 2008–2026

Manuscript reviewer (Psychometrika), 2012–2018

Manuscript reviewer (Journal of Educational Measurement), 2007–2019

Manuscript reviewer (Journal of Educational and Behavioral Statistics), 2015–2018

Manuscript reviewer (British Journal of Mathematical and Statistical Psychology), 2013–2014

Manuscript reviewer (Asia Pacific Education Review), 2010–2021

Manuscript reviewer (Educational Assessment), 2018

Manuscript reviewer (Educational and Psychological Measurement), 2011, 2013

Manuscript reviewer (International Journal of Testing), 2008–2015

Manuscript reviewer (Practical Assessment, Research, and Evaluation), 2021

Manuscript reviewer (EDUKACJA Quarterly), 2014–2015

Manuscript reviewer (SAGE Open), 2016

Manuscript reviewer (Educational Sciences: Theory & Practice), 2016

Manuscript reviewer (Journal of Educational Evaluation for Health Professions), 2018

Manuscript reviewer (Educational Measurement: Issues and Practice), 2018–2026

Manuscript reviewer (Applied Measurement in Education), 2019

Report reviewer (Education and Training Evaluation Commission Technical Report), 2018–2021

Proposal Review & Conference Service

Proposal reviewer (American Educational Research Association), 2010–2012

Proposal reviewer (International Association for Computerized Adaptive Testing), 2010

Proposal reviewer (National Council on Measurement in Education), 2007–2012

Proposal reviewer (Northeastern Educational Research Association), 2008–2009

Proposal reviewer (Association of Test Publishers), 2009–2019

Presentation discussant (National Council on Measurement in Education), 2009–2016

Session chair (International Test Commission), 2010

Session chair (International Association of Computerized Adaptive Testing), 2011

Industry mentor for Capstone Course at the Data Science and Analytics Master Program at Georgetown University, 2025

Leadership & Committee Service

Chair of Board of Directors, Chair of Foundation of Korean-American Educational Researchers Association (KAERA), 2021–present

NCME Jason Millman Award Committee Chair; Vice-chair, 2018–2020

NCME Annual Award Committee, 2017

President & Vice President of KAERA, 2015–2017

AERA Div. D Award Committee Chair/Vice-chair, 2015–2017

NCME Alicia Cascallar Award Committee, 2014–2015

Treasurer; Executive team (Korean-American Educational Researchers Association), 2012–2014

PUBLICATIONS (PEER-REVIEWED JOURNALS AND BOOK CHAPTERS)

Han, K. T., Lee, S.-H., & Malatesta, J. (2026). *Enhancing Test Taking Experience of Adaptive Tests with Review and Edit Features*. Manuscript submitted for revision to *Educational Measurement: Issues and Practice*.

Yuan, Y., & Han, K. T. (2026). *Adapting Bookmark Standard-Setting Method with LLMs for Item Difficulty Prediction*. Manuscript under review for publication.

Yuan, Y., & Han, K. T. (2026). *Modified Fade-Away Method for Conditional Item Exposure Control in CAT*. Manuscript under review for publication.

Lim, H., & Han, K. T. (2026). *IRT Residual-Based Approach to Detecting Item Parameter Drift in CAT*. Manuscript under review for publication.

Leng, D., Fu, Y., & Han, K. T. (2026). *The Edge Aversion Heuristic: Systematic Tendency in Rapid Guessing Behavior on High-Stakes Tests*. Manuscript under review for publication.

Han, K. T. (2025). Multistage testing with intersectional routing for short-length tests. In D. Yan, A. von Davier, & C. Lewis (Eds). *Research for practical issues and solutions in computerized multistage*. London, UK: Routledge.

Han, K. T., Dimitrov, D., & Al-Mashary, F. (2025). Multistage tests under D-scoring approach. In D. Yan, A. von Davier, & C. Lewis (Eds). *Research for practical issues and solutions in computerized multistage*. London, UK: Routledge.

Lim, H., Zhu, D., Choe, E. M., & Han, K. T. (2024). Detecting differential item functioning among multiple groups using IRT residual DIF framework. *Journal of Educational Measurement*. https://doi.org/10.1111/jedm.12415

Suk, Y., & Han, K. T. (2024). Evaluating intersectional fairness in algorithmic decision making using intersectional differential algorithmic functioning. *Journal of Educational and Behavioral Statistics*. https://doi.org/10.3102/10769986241269820

Lee, C., & Han, K. T. (2024). Evaluating the effectiveness of the standard error of score estimation as a CAT termination criterion. *Journal of Computerized Adaptive Testing, 11*(2). https://doi.org/10.7333/2410-1102013

Lim, H., & Han, K. T. (2024). An automated item pool assembly framework for maximizing item utilization for CAT. *Educational Measurement: Issues and Practice, 43*, 39–51. https://doi.org/10.1111/emip.12589

Suk, Y., & Han, K. T. (2023). A psychometric framework for evaluating fairness in algorithmic decision making: Differential algorithmic functioning. *Journal of Educational and Behavioral Statistics*. https://doi.org/10.3102/10769986231171711

Lim, H., Choe, E. M., & Han, K. T. (2022). A residual-based differential item functioning detection framework in item response theory. *Journal of Educational Measurement, 59*(1), 80–104.

Choe, E. M., & Han, K. T. (2022). Constructing a robust score scale from IRT scores with informed boundaries. *Journal of Educational Measurement*. https://doi.org/10.1111/jedm.12307

Han, K. T. (2020). Framework for developing multistage testing with intersectional routing for short-length tests. *Applied Psychological Measurement, 44*(2), 87–102.

Liu, C., Han, K. T., & Li, J. (2019). Compromised item detection for computerized adaptive testing. *Frontiers in Psychology, 10*(829). https://doi.org/10.3389/fpsyg.2019.00829

Han, K. T., Dimitrov, D., & Al-Mashary, F. (2019). Developing multistage tests using D-scoring method. *Educational and Psychological Measurement, 79*(5), 988–1008. https://doi.org/10.1177/0013164419841428

Han, K. T. (2018). Conducting simulation studies for computerized adaptive testing using SimulCAT: An instructional piece. *Journal of Educational Evaluation for Health Professions, 15*(20). https://doi.org/10.3352/jeehp.2018.15.20

Han, K. T. (2018). Components of the item selection algorithm in computerized adaptive testing. *Journal of Educational Evaluation for Health Professions, 15*(7). https://doi.org/10.3352/jeehp.2018.15.7

Han, K. T. (2018). PARSCALE. *SAGE Encyclopedia of Educational Research, Measurement and Evaluation*.

Han, K. T. (2018). WinGen. In W. J. van der Linden & R. K. Hambleton (Eds.). *Handbook of item response theory: Models, statistical tools, and applications*. Boca Raton, FL: CRC Press.

Guo, F., Meng, H., Han, K. T., & Talento-Miller, E. (2016). GMAT integrated reasoning section: Design and development. *China Examinations, 295*(11), 3–12.

Han, K. T. (2016). Maximum likelihood score estimation method with fences for short-length tests and computerized adaptive tests. *Applied Psychological Measurement, 40*(4), 289–301.

Han, K. T., & Rudner, L. M. (2016). Decision consistency. In C. Wells & M. Faulkner-Bond (Eds). *Educational measurement: From foundation to future*. New York, NY: Guilford.

Han, K. T., Wells, C. S., & Hambleton, R. K. (2015). Effect of adjusting pseudo-guessing parameter estimates on test scaling when item parameter drift is present. *Practical Assessment, Research & Evaluation, 20*(16). https://doi.org/10.7275/jyyy-wp74

Deng, N., Anatchkova, M. D., Waring, M. E., Han, K. T., & Ware, J. E. (2015). Testing item response theory invariance of the standardized quality-of-life disease impact scale (QDIS) in acute coronary syndrome patients: Differential functioning of items and scale. *Quality of Life Research, 24*(8), 1809–1822.

American Council on Education (2015). Educational measurement. (Korean Society for Educational Evaluation, & Korean-American Educational Research Association, Trans.). Seoul: Korea. *Original work published 2006*.

Han, K. T., & Paek, I. (2014). A review of commercial software packages for multidimensional IRT modeling. *Applied Psychological Measurement, 38*(6), 486–498.

Han, K. T., & Guo, F. (2014). Impact of violation of the missing-at-random assumption on full-information maximum likelihood method in multidimensional adaptive testing. *Practical Assessment, Research & Evaluation, 19*(2).

Han, K. T., & Kosinski, M. (2014). Software tools for multistage testing simulation. In D. Yan, A. von Davier, & C. Lewis (Eds). *Computerized multistage testing: Theory and applications*. Boca Raton, FL: CRC Press.

Han, K. T., & Guo, F. (2014). Assembling optimal modules on-the-fly. In D. Yan, A. von Davier, & C. Lewis (Eds). *Computerized multistage testing: Theory and applications*. Boca Raton, FL: CRC Press.

Han, K. T. (2013). MSTGen: Simulated data generator for multistage testing. *Applied Psychological Measurement, 37*(8), 666–668.

Han, K. T. (2013). Item pocket approach to allow response review and change in computerized adaptive testing. *Applied Psychological Measurement, 37*(4), 259–275.

Paek, I., & Han, K. T. (2013). Software review: IRTPRO 2.1 for Windows. *Applied Psychological Measurement, 37*(3), 242–252.

Talento-Miller, E., Guo, F., & Han, K. T. (2013). Examining test speededness by native language. *International Journal of Testing, 13*(2), 89–104.

Deng, N., Han, K. T., & Hambleton, R. K. (2013). A review of DIMPACK V.1.0: Conditional covariance-based dimensionality analysis package. *Applied Psychological Measurement, 37*, 162–172.

Han, K. T. (2012). An efficiency balanced information criterion for item selection in computerized adaptive testing. *Journal of Educational Measurement, 49*(3), 225–246.

Han, K. T., Wells, C. S., & Sireci, S. G. (2012). Impact of multidirectional item parameter drift on test equating and proficiency estimates. *Applied Measurement in Education, 25*(2), 97–117.

Han, K. T. (2012). SimulCAT: Windows software for simulating computerized adaptive test administration. *Applied Psychological Measurement, 36*(1), 64–66.

Han, K. T. (2012). Fixing the c parameter in the three-parameter logistic model. *Practical Assessment, Research & Evaluation, 17*(1).

Han, K. T. (2009). IRTEQ: Windows application that implements IRT scaling and equating. *Applied Psychological Measurement, 33*(6), 491–493.

Liang, T., Han, K. T., & Hambleton, R. K. (2009). ResidPlots-2: Computer software for IRT graphical residual analyses. *Applied Psychological Measurement, 33*(5), 411–412.

Sireci, S. G., Han, K. T., & Wells, C. S. (2008). Methods for evaluating the validity of test scores for English language learners. *Educational Assessment, 13*(2), 108–131.

Sireci, S. G., Wells, C. S., & Han, K. T. (2008). Identifying item parameter drift in multistage adaptive testing. *CLEAR Exam Review, 19*(1), 14–21.

Han, K. T. (2007). WinGen: Windows software that generates IRT parameters and item responses. *Applied Psychological Measurement, 31*, 457–459.

SELECTED PRESENTATIONS & REPORTS

Shen, Y., He, Q., & Han, K. T. (2026, April). Examining nonresponse patterns in GMAT registration data and their associations with test performance. Presentation will be delivered at the annual meeting of AERA, Los Angeles.

Han, K. T. (2026, April). Section time limits don't control item-level speededness. Presentation will be delivered at the annual meeting of AERA, Los Angeles.

Yuan, Y., & Han, K. T. (2026, April). Adopting bookmark standard-setting method with LLMs for item difficulty prediction. Paper will be presented at the annual conference of NCME, Los Angeles.

Lee, S., & Han, K. T. (2026, April). Implementing multistage testing using existing computerized adaptive testing system with minimal changes. Paper will be presented at the annual conference of NCME, Los Angeles.

Lim, H., Choe, E., & Han, K. T. (2026, April). Global maximum is not always best in 3PL maximum-likelihood scoring: Smarter initialization. Paper will be presented at the annual conference of NCME, Los Angeles.

Han, K. T. (2026, April). Section time limits don't control item-level speededness: A local independence problem. Presentation will be delivered at the annual conference of NCME, Los Angeles.

Zhang, Y., Fu, Y., & Han, K. T. (2026, April). AI-powered prediction of 2PLM parameters from item content. Paper will be presented at the annual conference of NCME, Los Angeles.

Yuan, Y., & Han, K. T. (2025, November). Localized fade-away method for controlling item exposure across performance levels. Presentation delivered at the conference on Test Security, Amherst, MA.

Lim, H., & Han, K. T. (2025, April). IRT residual-based approach to evaluating item parameter drift in CAT. Paper presented at the annual conference of NCME, Denver.

Malatesta, J., & Han, K. T. (2025, April). Allowing answer changes in computerized adaptive tests: Yes, we can! Paper presented at the annual conference of NCME, Denver.

Malatesta, J., Satkus, P., & Han, K. T. (2025, April). Rescaling the GMAT. Presentation delivered at the annual conference of NCME, Denver.

Leng, D., Fu, Y., & Han, K. T. (2025, April). Understanding the behavior and implications of rapid guessing. Paper presented at the annual conference of NCME, Denver.

Yuan, Y., & Han, K. T. (2025, April). Modified fade-away method for conditional item exposure control in CAT. Paper presented at the annual conference of NCME, Denver.

Satkus, P., & Han, K. T. (2025, April). What do test-takers want? Examining the difficulty of CAT. Paper presented at the annual conference of NCME, Denver.

Lee, S., & Han, K. T. (2025, April). Adaptively regulated interim theta bounds for optimizing CAT item selection. Poster presented at the annual conference of NCME, Denver.

Han, K. T. (2024, September). Enhancing measurement efficiency of MST by making routing stage adaptive. Presentation delivered at biennial meeting of IACAT, Seoul, Korea.

Han, K. T., Lim, H., & Lee, Y. (2024, September). Development of score scale for MST-based Sejong Korean Assessment. Presentation delivered at biennial meeting of IACAT, Seoul, Korea.

Kim, J. P., Demirkaya, O., & Han, K. T. (2024, September). Optimal multistage adaptive testing shaping modules with various shaping rules. Presentation delivered at biennial meeting of IACAT, Seoul, Korea.

Satkus, P., Fu, Y., & Han, K. T. (2024, July). Advancing enemy item detection using transformer-augmented NLP. Presentation delivered at IMPS, Prague, Czech Republic.

Lee, S-H., & Han, K. T. (2024, July). Adaptively regulated interim theta bounds for optimizing CAT item selection. Presentation delivered at IMPS, Prague, Czech Republic.

Wang, B., Lim, H., & Han, K. T. (2024, April). Detection of differential item functioning in multidimensional testing using IRT-residual DIF framework. Presentation delivered at NCME, Philadelphia.

Fu, Y., Satkus, P., & Han, K. T. (2024, April). Evaluating the performance of GPT-4 on the GMAT exam questions. Paper presented at NCME, Philadelphia.

Lim, H., & Han, K. T. (2024, April). Streamlining test pool production using Honeycomb framework. Presentation delivered at NCME, Philadelphia.

Han, K. T., Fu, Y., & Lee, S. (2024, April). New CAT design for optimizing measurement efficiency and test security. Presentation delivered at NCME, Philadelphia.

Satkus, P., & Han, K. T. (2024, April). Improving test-taking experience of CAT with review & edit feature. Presentation delivered at NCME, Philadelphia.

Fu, Y., & Han, K. T. (2024, April). Conducting triangulated predictive validity study using test data from past. Presentation delivered at NCME, Philadelphia.

Malatesta, J., Lim, H., & Han, K. T. (2024, April). Designing score scales and student reports in presence of test construct changes. Presentation delivered at NCME, Philadelphia.

Suk, Y., & Han, K. T. (2024, April). A framework for evaluating intersectional fairness in algorithmic decision making. Paper presented at NCME, Philadelphia.

Lim, H., & Han, K. T. (2023, July). A new approach to evaluating item parameter drift in computerized adaptive testing. Paper presented at IMPS, College Park, MD.

Lim, H., & Han, K. T. (2023, April). Automated test assembly in operational assessment programs. Paper presented at NCME, Chicago.

Lim, H., Zhu, D., Choe, E., & Han, K. T. (2023, April). Detecting differential item functioning among multiple groups using IRT residual DIF framework. Paper presented at NCME, Chicago.

Fu, Y., & Han, K. T. (2023, April). Enemy item detection for quantitative item type. Paper presented at NCME, Chicago.

Suk, Y., & Han, K. T. (2023, April). A framework for evaluating fairness in algorithmic decision making: Differential algorithmic functioning. Paper presented at NCME, Chicago.

Zhu, D., Fu, Y., & Han, K. T. (2023, April). Understanding factors for creating isomorphic instances in automated item generation. Paper presented at NCME, Chicago.

Lee, S., Han, K. T., & Satkus, P. (2023, March). Improving test fairness in CAT with equitable test latency. Presentation delivered at the ATP Innovations in Testing Conference, Dallas, Texas.

Satkus, P., & Han, K. T. (2022, September). Considering test-taking experience for item selection in CAT. Paper presented at the biennial conference of IACAT, Frankfurt, Germany.

Lim, H., & Han, K. T. (2022, April). Factors affecting item calibration using adaptively administered test data. Poster presented at IMPS, Bologna, Italy.

Lim, H., Choe, E. M., & Han, K. T. (2022, April). Detecting differential item functioning in CAT using IRT residual DIF approach. Paper presented at NCME, San Diego.

Fu, Y., & Han, K. T. (2022, April). Enemy item detection for different item types. Paper presented at NCME, San Diego.

Lim, H., Choe, E. M., Han, K. T., Lee, S., & Hong, M. (2021, June). IRT residual approach to detecting differential item functioning. Paper presented at NCME, Online.

Hong, M., Fu, Y., & Han, K. T. (2021, June). Prediction of item difficulty using natural language processing with topic modeling. Paper presented at NCME, Online.

Pan, Y., Lee, S., & Han, K. T. (2021, June). Understanding different ways to compute measurement errors and score reliability of adaptive tests. Paper presented at NCME, Online.

Li, J., Liu, C., & Han, K. T. (2020, September). Compromised item detection for computerized adaptive testing. Presentation delivered at the annual meeting of ATP.

Fu, Y., & Han, K. T. (2020, July). Use language models to solve sentence-correction multiple-choice questions. Paper presented at IMPS, College Park, MD.

Pan, Y., Lee, S., & Han, K. T. (2020). Understanding different ways to compute measurement errors and score reliability for adaptive tests. Paper accepted at NCME, San Francisco, CA.

Fu, Y., & Han, K. T. (2019, June). A comparison of three approaches that permit answer review in CAT. Paper presented at IACAT, Twin City, MN.

Choe, E., & Han, K. T. (2019, April). Determining optimal bounds of IRT scores for developing a meaningful score scale. Paper delivered at NCME, Toronto.

Shi, Y., & Han, K. T. (2019, April). Response time processes in computerized adaptive testing. Paper delivered at NCME, Toronto.

Shi, Y., De Boeck, P., & Han, K. T. (2019, April). Heterogeneity of response time on task effect in computerized adaptive testing. Paper delivered at NCME, Toronto.

Han, K. T. (2019, March). Multistage testing: Theory and practice. Presentation delivered at ATP, Orlando, FL.

Han, K. T. (2018, September). Options to allow test takers to review and change responses in CAT. Presentation delivered at European Association of Test Publishers, Athens, Greece.

Han, K. T. (2018, February). Executive assessment based on multistage testing with cross-sectional routing. Presentation delivered at ATP, San Antonio, TX.

Han, K. T. (2017, April). Executive assessment based on multistage testing with cross-sectional routing. Paper presented at AERA, San Antonio, TX.

Meng, H., Han, K. T., & Guo, F. (2017, April). Using new items operationally in adaptive tests. Paper presented at AERA, San Antonio, TX.

Lee, C., & Han, K. T. (2017, April). Evaluating effectiveness of standard error of score estimation as a CAT termination criterion. Paper presented at AERA, San Antonio, TX.

Meng, H., Han, K. T., & Guo, F. (2016, July). Conquer the IRT hurdle for adaptive test designs. Presentation delivered at International Test Commission, Vancouver, BC.

Han, K. T. (2015, September). Happy CAT: Options to allow test takers to review and change responses in CAT. Keynote speech at IACAT, Cambridge, UK.

Han, K. T., Guo, F., & Talento-Miller, E. (2015, September). Multistage testing with routing based on performance on different test sections. Paper presented at IACAT, Cambridge, UK.

Lee, C., & Han, K. T. (2015, September). Evaluating effectiveness of standard error of score estimation as a termination criterion in CAT. Paper presented at IACAT, Cambridge, UK.

Han, K. T. (2015, July). Maximum likelihood score estimation method for short-length tests and computerized adaptive tests. Paper presented at IMPS, Beijing, China.

Han, K. T., & Rudner, L. M. (2014). Item pool construction using mixed integer quadratic programming (MIQP). GMAC Research Report RR-14-01, June 9, 2014.

Han, K. T., & Paek, I. (2014, July). Comparison of software for uni- and multi-dimensional IRT model estimation. Paper presented at ITC, San Sebastian, Spain.

Han, K. T. (2014, July). Evaluating relationship among subtests using structural equation modeling. Presentation delivered at ITC, San Sebastian, Spain.

Han, K. T. (2014, April). Sharing research and beyond. Presentation delivered at NCME, Philadelphia, PA.

Han, K. T., & Paek, I. (2014, April). A comparison of commercial software for multidimensional IRT modeling. Paper presented at NCME, Philadelphia, PA.

Han, K. T. (2014, March). Using structural equation modeling approach to investigate latent factor structure across subtests. Presentation delivered at ATP, Scottsdale, Arizona.

Kim, H., Park, J.-I., Park, E. A., Seo, M., & Han, K. T. (2013). Comprehensive improvement planning of assessment for enhancing creativity and character education. KICE Research Report RRE 2013-20, December 31, 2013.

Han, K. T. (2013, April). Impact of violation of the missing-at-random assumption on item calibration and recalibration for multidimensional adaptive testing. Paper presented at AERA, San Francisco, CA.

Han, K. T. (2013, April). Evaluating item parameter drift in computerized adaptive testing. Presentation delivered at NCME, San Francisco, CA.

Han, K. T., & Guo, F. (2013). An approach to assembling optimal multistage testing modules on-the-fly. GMAC Research Report RR-13-01, March 4, 2013.

Han, K. T. (2013, February). A series of computer software tools for test simulations. Presentation delivered at ATP, Hollywood, FL.

Han, K. T., & Guo, F. (2012, August). Multistage testing by shaping a module at each stage. Presentation delivered at IACAT, Sydney, Australia.

Han, K. T. (2012, August). Simulation tools for various computerized adaptive tests. Presentation delivered at IACAT, Sydney, Australia.

Talento-Miller, E., Han, K. T., & Guo, F. (2012, August). Impact of pre-test item position on calibration and ability estimation. Paper presented at IACAT, Sydney, Australia.

Han, K. T., & Rudner, L. M. (2012, July). Free tools for learning and applying item response theory. Presentation delivered at ITC, Amsterdam, Netherlands.

Han, K. T., & Guo, F. (2012, July). Robustness of the full-information maximum likelihood method for adaptively administered test data. Presentation delivered at ITC, Amsterdam, Netherlands.

Han, K. T. (2012, April). A new item selection criterion to incorporate item efficiency level in computerized adaptive testing. Paper presented at NCME, Vancouver, BC.

Talento-Miller, E., Han, K. T., & Guo, F. (2012, April). Analysis of embedded pretest items by position. Paper presented at NCME, Vancouver, BC.

Han, K. T., Rudner, L. M., Guo, F., & Talento-Miller, E. (2012, February). Detecting item parameter drift in computerized adaptive testing. Presentation delivered at ATP, Palm Spring, CA.

Han, K. T., Guo, F., Talento-Miller, E., & Rudner, L. M. (2011, October). Recalibrating items using adaptively administered response data. Paper presented at IACAT, Pacific Grove, CA.

Han, K. T. (2011, April). Item pocket approach to allow response review and change in computerized adaptive testing. Paper presented at NCME, New Orleans, LA.

Wells, C. S., Keller, L., & Han, K. T. (2011, April). Minimizing the effect of model misfit on the invariance property for the three-parameter logistic model. Paper presented at NCME, New Orleans, LA.

Talento-Miller, E., Guo, F., & Han, K. T. (2011, April). Examining test speededness by native language. Paper presented at NCME, New Orleans, LA.

Talento-Miller, E., Han, K. T., & Guo, F. (2011). Guess again: The effect of correct guesses on scores in an operational CAT program. GMAC Research Report RR-11-04, March 14, 2011.

Han, K. T. (2011, March). Identifying item needs for item pool construction using the mixed integer programming. Presentation delivered at ATP, Phoenix, AZ.

Han, K. T., & Guo, F. (2011). Potential impact of item parameter drift due to practice and curriculum change on item calibration in computerized adaptive testing. GMAC Research Report RR-11-02, January 1, 2011.

Han, K. T. (2010, July). Discussion of three-parameter logistic model: What is the best guess at the guessing parameter? Paper presented at ITC, Hong Kong, China.

Han, K. T., & Guo, F. (2010, July). Impact of compromised items on item pool calibration. Paper presented at ITC, Hong Kong, China.

Talento-Miller, E., Guo, F., & Han, K. T. (2010, July). Examining test speededness by native language. Paper presented at ITC, Hong Kong, China.

Talento-Miller, E., Han, K. T., & Guo, F. (2010, June). The effect of correct guesses on scores in an operational CAT program. Paper presented at IACAT, Arnhem, Netherlands.

Han, K. T. (2010, May). Comparison of non-Fisher-information item selection criteria in fixed-length CAT. Paper presented at NCME, Denver, CO.

Han, K. T., & Rudner, L. M. (2010, May). Implementing the mixed integer programming for constructing item pools for CAT. Paper presented at NCME, Denver, CO.

Talento-Miller, E., Han, K. T., & Guo, F. (2010, April). Mythbusters: A practical look at the effect of position of correct responses on a computerized adaptive test. Paper presented at NCME, Denver, CO.

Han, K. T. (2010, February). Evaluating efficiency of item selection methods and item needs. Presentation delivered at ATP, Orlando, FL.

Han, K. T. (2009, June). Gradual maximum information ratio approach to item selection in computerized adaptive testing. GMAC Research Report RR-09-07, June 25, 2009.

Han, K. T., Wells, C. S., & Hambleton, R. K. (2009, July). Impact of item parameter drift on pseudo-guessing parameter estimates and test equating. Paper presented at the Psychometric Society, Cambridge, UK.

Han, K. T. (2009, June). Gradual maximum information ratio approach to item selection in computerized adaptive testing. Paper presented at GMAC 2009 CAT Conference, Minneapolis, MN.

Kim, J., & Han, K. T. (2009, April). New approaches to test equating when construct response items are used as linking items. Presentation delivered at NCME, San Diego, CA.

Han, K. T., & Wells, C. S. (2008, July). Impact of multidirectional item parameter drift on test equating and proficiency estimates. Paper presented at the Psychometric Society, Durham, NH.

Lee, J. H., & Han, K. T. (2008, July). Optimizing the number of items for each module in multi-stage testing. Presentation delivered at the Psychometric Society, Durham, NH.

Zenisky, A. L., Han, K. T., Yoo, H., Smith, Z. R., & Patton, J. (2008). Evaluation of the CATs content balancing algorithm (Center for Educational Assessment Report No. 682). Amherst, MA: UMass.

Sireci, S. G., Baldwin, P., Martone, D., Laguilles, A. Z., Lam, W., Lewis, C., Han, K. T., & Hambleton, R. K. (2008). Massachusetts adult proficiency tests technical manual: Version 2 (CEA Report No. 677). Amherst, MA: UMass.

Liang, T., Han, K. T., & Baldwin, P. (2008, March). Computer software for graphical residual analysis. Poster presented at NCME, New York, NY.

Liang, T., Han, K. T., & Hambleton, R. K. (2008). ResidPlots: Computer program for data-model fit and residual analysis [computer program]. Amherst, MA: UMass.

Han, K. T., Xu, X., & von Davier, M. (2008, January). Investigating the dimensionality of NAEP reading data using the general diagnostic model. Paper presented at the International Conference on Learning Competency, Seoul, Korea.

Han, K. T. (2008, January). IRT test equating procedures using IRTEQ. Paper presented at the International Conference on Learning Competency, Seoul, Korea.

Han, K. T., & Sireci, S. G. (2007, October). Statistical models for evaluating educational gain at the program level for adult basic education programs. Paper presented at NERA, Rocky Hill, CT.

Liang, T., Sireci, S. G., Wells, C. S., & Han, K. T. (2007, October). Evaluating item parameter drift in computerized adaptive testing. Paper presented at NERA, Rocky Hill, CT.

Han, K. T., & Hambleton, R. K. (2007). User's manual: WinGen (CEA Report No. 642). Amherst, MA: UMass.

Han, K. T. (2007). HSGen: Hierarchical Score Generator [computer program]. Amherst, MA: UMass.

Han, K. T. (2007). Using 3-level hierarchical linear model for evaluating educational gain at the program level (CEA Report No. 628). Amherst, MA: UMass.

Han, K. T. (2007). WinGen2: Windows software that generates IRT parameters and item responses [computer program]. Amherst, MA: UMass.

Sireci, S. G., Wells, C. S., Baldwin, P., & Han, K. T. (2007, April). Detection of item degradation: Evaluating item parameter drift in computerized-adaptive testing. Paper presented at SIOP, New York, NY.

Sireci, S. G., & Han, K. T. (2007, April). Methods for evaluating the validity of test scores for English language learners. Paper presented at AERA, Chicago, IL.

Han, K. T., & Wells, C. S. (2007, April). Impact of differential item functioning (DIF) on test equating and proficiency estimates. Paper presented at NCME, Chicago, IL.

Sireci, S. G., Baldwin, P., Martone, D., & Han, K. T. (2007, April). Determining cutpoints on a multi-stage test for federally established proficiency levels. Paper presented at NCME, Chicago, IL.

Han, K. T., Baldwin, P., & Sireci, S. G. (2006). Analyses of differential item functioning of potential MAPT items (CEA Report No. 616). Amherst, MA: UMass.

Han, K. T. (2006). Establishing educational gain levels on newly developed assessments for ABE (CEA Report No. 615). Amherst, MA: UMass.

Sireci, S. G., Baldwin, P., Martone, D., Laguilles, A. Z., Hambleton, R. K., & Han, K. T. (2006). Massachusetts adult proficiency tests technical manual (CEA Report No. 600). Amherst, MA: UMass.

Han, K. T. (2006, October). WinGen: Windows software that generates IRT parameters and responses. Presentation delivered at NERA, Kerhonkson, NY.

Han, K. T. (2006, June). No such thing as content validity: Is construct validity a safe zone? Paper presented at the Psychometric Society, Montreal, Canada.

Han, K. T., Sireci, S. G., Wells, C. S., & Zenisky, A. L. (2006, April). Methods for evaluating educational gains at the program level. Paper presented at NCME, San Francisco, CA.

Han, K. T., Sireci, S. G., Wells, C. S., & Zenisky, A. L. (2005, October). Methods for evaluating educational gains at the program level. Paper presented at NERA, Kerhonkson, NY.

Han, K. T. (2005, April). A confirmatory factor analysis of the Revised Protestant Work Ethic Scale. Poster presented at NEERO, Northampton, MA.

Sireci, S. G., Wells, C. S., Han, K. T., & Zenisky, A. L. (2005). Evaluating educational gains at the program level (CEA Report No. 559). Amherst, MA: UMass.

INVITED KEYNOTES, SEMINARS, LECTURES, & WORKSHOPS

Han, K. T. (2026, March). Transforming education through the integration of (adaptive) learning and assessment for the age of AI [Invited talk]. The 1st International Conference: Toward Structural Innovation in Education for the Age of AI, Yonsei University, Seoul, South Korea.

Han, K. T. (2025, May). Practices and trends in modern test development [Invited lecture]. Department of Education, College of Education, Ewha Womans University, Seoul, South Korea.

Han, K. T. (2024, September). Sequel to Happy CAT from 10 years ago: Implementation of the review and edit feature in a high-stakes application of CAT. Keynote speech at IACAT, Seoul, Korea.

Han, K. T. (2024, September). Innovative CAT designs for improved testing experience & learning. Keynote speech at the 9th international conference on measurement and evaluation in education and psychology, Eskisehir, Türkiye.

Yan, D., von Davier, A., & Han, K. T. (2024, April). Computerized multistage adaptive testing. Pre-conference workshop at NCME, Philadelphia.

Han, K. T. (2023, May). Computerized adaptive testing for the future of psychological assessments. Invited lecture at Korea Institute of Behavioral Sciences.

Han, K. T. (2023, April). Latest trends in the educational test industry. Invited speech at the 40-year anniversary conference of Korean Society for Educational Evaluation at KICE.

Han, K. T. (2023, April). The future of psychometrics and test development with AI. Invited lecture at Korea Institute of Curriculum and Evaluation.

Han, K. T. (2022, October). Early considerations for computer adaptive testing for standardized assessment in Africa. Invited speech at the annual meeting of Association of Computer Adaptive Testing in Africa.

Han, K. T. (2022, March). Multistage testing simulations using D-scoring framework. 3-day workshop at National Center for Assessment in Saudi Arabia.

Zenisky, A., & Han, K. T. (2021, January). Multistage testing course. 3-week workshop organized by World Bank.

Han, K. T. (2020, July). Computerized adaptive testing: Theories and practices. Workshop at the Korean Society for Educational Evaluation meeting.

Yoo, H., Oh, H.-J., & Han, K. T. (2019, April). Software packages for IRT-based test simulation: WinGen3, SimulCAT, MSTGen, and IRTEQ. Pre-conference workshop at NCME, Toronto.

Yan, D., von Davier, A., & Han, K. T. (2019, April). Computerized multistage adaptive testing. Pre-conference workshop at NCME, Toronto.

Lee, W.-C., Kim, H., & Han, K. T. (2018, July). Tools for equating. Pre-conference workshop at ITC, Montreal, Canada.

Han, K. T. (2018, April). Measurement problems – a look back to help us look ahead. Presentation at NCME, New York.

Yan, D., von Davier, A., & Han, K. T. (2018, April). Computerized multistage adaptive testing. Pre-conference workshop at NCME, New York.

Han, K. T. (2017, April). Recent innovations in adaptive testing. Presentation at AERA, San Antonio, TX.

Yan, D., von Davier, A., & Han, K. T. (2017, April). Computerized multistage adaptive testing. Pre-conference workshop at NCME, San Antonio, TX.

Han, K. T. (2016, October). Recent changes around self-regulated learning and future role of textbooks. Keynote at the annual International Textbook Symposium, Seoul, South Korea.

Han, K. T. (2016, April). Presidential Address at KAERA, Washington, DC.

Yan, D., von Davier, A., & Han, K. T. (2016, April). Computerized multistage adaptive testing. Pre-conference workshop at NCME, Washington, DC.

Han, K. T. (2015, September). Happy CAT: Options to allow test takers to review and change responses in CAT. Keynote speech at IACAT, Cambridge, UK.

Yan, D., von Davier, A., & Han, K. T. (2015, September). Computerized multistage adaptive testing. Pre-conference workshop at IACAT, Cambridge, UK.

Han, K. T. (2015, August). The new big data: Human-centered analytics. Webinar at Georgetown University Alumni Association.

Han, K. T. (2015, July). Multistage testing. Pre-conference workshop at IMPS, Beijing, China.

Sireci, S. G., & Han, K. T. (2015, April). Contemporary problems in educational measurement (satirical session). Presentation at NCME, Chicago, IL.

Han, K. T. (2014, December). Educational research and measurement: Past, present, and future. Keynote at Krakow International Conference on Educational Research and Measurement, Krakow, Poland.

Han, K. T. (2014, October). Multistage testing. Pre-conference workshop at IACAT.

Han, K. T. (2014, August). Fundamentals of multidimensional item response theory. Seminar at Sung-Kyun-Kwan University, Seoul, South Korea.

Han, K. T. (2014, November). Computerized adaptive testing. 20-hour coursework at Jagiellonian University, Krakow, Poland. Funded by Educational Research Institute.

Han, K. T. (2013, May). Computerized adaptive testing. 20-hour coursework at Jagiellonian University, Krakow, Poland. Funded by Educational Research Institute.

Han, K. T. (2012, November). Computerized adaptive testing. Workshop at Teach Your Children Well conference, UMass Amherst.

Han, K. T. (2012, August). Applications of IRT. Seminar at Sung-Kyun-Kwan University, Seoul, South Korea.

Han, K. T. (2010, July). Practical issues of three parameter IRT model. Seminar at Yonsei University, Seoul, South Korea. Funded by Brain Korea 21 project.

Han, K. T. (2010, July). Computer programming for psychometric research. Seminar at Sung-Kyun-Kwan University, Seoul, South Korea. Funded by Brain Korea 21 project.

Han, K. T. (2009, May). Fundamentals of computerized adaptive testing. Seminar at Sung-Kyun-Kwan University, Seoul, South Korea. Funded by Brain Korea 21 project.

Han, K. T. (2008, January). Monte-Carlo simulations for IRT-based test development. 5-day workshop at Sung-Kyun-Kwan University, Seoul, South Korea. Funded by Brain Korea 21 project.

Han, K. T. (2007, January). Fundamentals of IRT. 5-day workshop at Sung-Kyun-Kwan University, Seoul, South Korea. Funded by Brain Korea 21 project.

PROFESSIONAL AFFILIATIONS

National Council on Measurement in Education (NCME)

Psychometric Society

International Association for Computerized Adaptive Testing (IACAT)

American Educational Research Association (AERA)

Korean-American Educational Researchers Association (KAERA)

International Testing Commission (ITC)

Association of Test Publishers (ATP)

KEY DOCTORAL COURSEWORK

Multidimensional Scaling by Prof. Stephen G. Sireci

Principles of Testing by Prof. Ronald K. Hambleton

Item Response Theory I, II by Prof. Ronald K. Hambleton

Multivariate Statistics I, II by Prof. Lisa Keller

Advanced Validity Theory by Prof. Stephen G. Sireci

Structural Equation Modeling by Prof. Craig S. Wells

Classical Testing Theory by Prof. Ronald K. Hambleton

Test Construction by Prof. Stephen G. Sireci

Fortran / S Plus Programming by Prof. Lisa Keller and Prof. Craig S. Wells

Hierarchical Linear Model by Prof. Aline G. Sayer

Non-parametric statistical analysis by Prof. Craig S. Wells

Educational Research Methods by Prof. Ronald K. Hambleton

COMPUTER LANGUAGES / SOFTWARE PACKAGES

Claude Cowork; Claude Code

OpenAI LLM Platform; Ollama platform

Python

Microsoft Visual Studio .NET (C#.NET & Visual Basic .NET)

Linear / Mixed Integer Programming

Fortran 90/95

IRT software packages including PARSCALE, Winsteps, & FlexMIRT

SAS, SPSS, S plus / R

SEM software packages (Mplus, Lisrel, AMOS)

HLM

HTML / CSS / JavaScript

OTHER ACTIVITIES

Virginia State Indoor Archery championship 1st place recurve/male/senior, 2022

USA National Indoor Archery 2017, 3rd place recurve/male/senior in VA

USA National Indoor Archery 2015, 1st place male/senior/recurve (guest)

Virginia State Indoor Archery championship 1st place recurve/male/senior, 2015

USA National Indoor Archery 2014, 1st place male/senior/recurve (guest)

USA Archery Level 2 Certified Coach